

Parent Functions IV: b^x and log base b**TEACHER KEY** | Algebra II | Unit 1 | Day 7 | TEK A2.2.A**Objective:** I can graph exponential and logarithmic parent functions for $b = 2, 10$, and e , analyze their key attributes, and notice that their domains and ranges trade places.**Vocabulary****Exponential function** — the variable is in the **exponent**, as in $f(x) = 2^x$.**Base** — the number being raised to a power. We use $b = 2, 10$, and **$e \sim 2.718$** .**Logarithm** — log base b of a asks: what **exponent** on b produces a?**Natural log** — **$\ln(x)$** means log base e . **Common log** — **$\log(x)$** means log base 10.**Part 1 — The Exponential Function, $f(x) = 2^x$** *Fill in the table. Negative exponents give fractions, never negative outputs.*

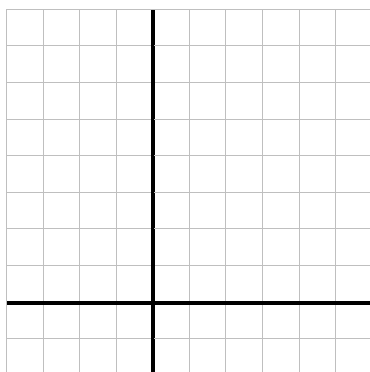
x	-2	-1	0	1	2	3
$f(x) = 2^x$	1/4	1/2	1	2	4	8

Can 2^x ever equal 0 or a negative number? **no - the output is always positive**So there is a horizontal asymptote at **$y = 0$** and a y-intercept at **$(0, 1)$** .**Part 2 — The Logarithmic Function, $f(x) = \log_2(x)$** *This undoes the exponential. Compare it to your Part 1 table.*

x	1/4	1/2	1	2	4	8
$f(x) = \log_2(x)$	-2	-1	0	1	2	3

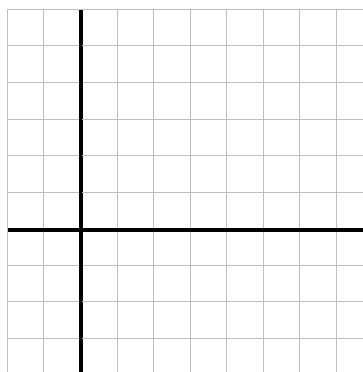
What happened to the table from Part 1? **the x and y values traded places**What is $\log_2(0)$? **undefined - no power of 2 gives 0**So there is a vertical asymptote at **$x = 0$** and an x-intercept at **$(1, 0)$** .**Part 3 — Graph Them Both***Each square is 1 unit. Sketch the asymptote as a dashed line first, then plot.*

A. $f(x) = 2^x$



Asymptote: $y = 0$

B. $f(x) = \log_2(x)$



Asymptote: $x = 0$

Part 4 — Compare the Key Attributes

Attribute	$f(x) = b^x$	$f(x) = \log \text{ base } b$
Domain (interval)	$(-\infty, \infty)$	$(0, \infty)$
Range (interval)	$(0, \infty)$	$(-\infty, \infty)$
Intercept	$(0, 1)$	$(1, 0)$
Asymptote	horizontal, $y = 0$	vertical, $x = 0$
Increasing / decreasing	increasing	increasing

Look at the first two rows. The domain and range have **traded places**, and so did the intercepts and asymptotes. That is a strong hint these are **inverses** - Day 10.

Part 5 — Changing the Base

All bases greater than 1 give the same shape. Only the steepness changes.

x	0	1	2
2^x	1	2	4

x	0	1	2
10^x	1	10	100

Which grows faster, and why? **10^x - a bigger base multiplies by more each step**

What do all three exponential graphs share? **the y-intercept $(0, 1)$**

Exit Ticket

A function has domain $(0, \infty)$, range $(-\infty, \infty)$, an x-intercept at $(1, 0)$, and a vertical asymptote at $x = 0$.

Name the family: **logarithmic** Name its inverse family: **exponential**